DEPARTMENT OF ELECTRICAL & ELECTRONIC ENGINEERING

Experiment No. : 1

Experiment Name: Introduction to MATLAB & Simulink as Control System I lab software

OBJECTIVES

1. **Understanding the basic commands based on built in example.**
2. **Getting used to with Simulink blocks for system development.**

PRELAB WORK:

1. **Read this laboratory tutorial carefully before coming to the laboratory class, so that you know what is required.**
2. Try to follow the lectures of EEE 4701.
3. Familiarize yourself with relevant MATLAB functions and codes necessary for this experiment.

Objective-1: **Understanding the basic commands based on built in example.**

MATLAB Commands List

The following list of commands can be very useful for future reference. Use "help" in MATLAB for more information on how to use the commands.

In these tutorials, we use commands both from MATLAB and from the Control Systems Toolbox, as well as some commands/functions which we wrote ourselves. For those commands/functions which are not standard in MATLAB, we give links to their descriptions.

Note: MATLAB commands from the control system toolbox are highlighted in red. Non-standard MATLAB commands are highlighted in green.

Command	Description
abs	Absolute value
acker	Compute the K matrix to place the poles of A-BK, see also place
axis	Set the scale of the current plot, see also plot, figure
bode	Draw the Bode plot, see also logspace, margin, nyquist1
c2d	Continuous system to discrete system
clf	Clear figure
conv	Convolution (useful for multiplying polynomials), see also deconv
ctrb	The controllability matrix, see also obsv
deconv	Deconvolution and polynomial division, see also conv
det	Find the determinant of a matrix
dlqr	Linear-quadratic regulator design for discrete-time systems, see also lqr
eig	Compute the eigenvalues of a matrix
eps	MATLAB's numerical tolerance
feedback	Connect linear systems in a feedback loop
figure	Create a new figure or redefine the current figure, see also subplot, axis
for	For, next loop
format	Number format (significant digits, exponents)
function	Creates function m-files
grid	Draw the grid lines on the current plot
gtext	Add a piece of text to the current plot, see also text
help	HELP!
hold	Hold the current graph, see also figure
if	Conditionally execute statements
imag	Returns the imaginary part of a complex number, see also real
impulse	Impulse response of linear systems, see also step, lsim
input	Prompt for user input

inv	Find the inverse of a matrix
legend	Graph legend
length	Length of a vector, see also size
linspace	Returns a linearly spaced vector
lnyquist	Produce a Nyquist plot on a logarithmic scale, see also nyquist1
log	natural logarithm, also log10: common logarithm
loglog	Plot using log-log scale, also semilogx/semilogy
logspace	Returns a logarithmically spaced vector
lqr	Linear quadratic regulator design for continuous systems, see also dlqr
lsim	Simulate a linear system, see also step, impulse
margin	Returns the gain margin, phase margin, and crossover frequencies, see also bode
minreal	Produces a minimal realization of a system (forces pole/zero cancellations)
norm	Norm of a vector
nyquist1	Draw the Nyquist plot, see also lnyquist. Note this command was written to replace the MATLAB standard command nyquist to get more accurate Nyquist plots.
obsv	The observability matrix, see also ctrb
ones	Returns a vector or matrix of ones, see also zeros
place	Compute the K matrix to place the poles of A-BK, see also acker
plot	Draw a plot, see also figure, axis, subplot.
poly	Returns the characteristic polynomial
polyval	Polynomial evaluation
print	Print the current plot (to a printer or postscript file)
pzmap	Pole-zero map of linear systems
rank	Find the number of linearly independent rows or columns of a matrix
real	Returns the real part of a complex number, see also imag
rlocfind	Find the value of k and the poles at the selected point
rlocus	Draw the root locus
roots	Find the roots of a polynomial
rscale	Find the scale factor for a full-state feedback system
set	Set(gca,'Xtick',xticks,'Ytick',yticks) to control the number and spacing of tick marks on the axes
sgrid	Generate grid lines of constant damping ratio (zeta) and natural frequency (Wn), see also sigrid, zgrid
sigrid	Generate grid lines of constant settling time (sigma), see also sgrid, zgrid
size	Gives the dimension of a vector or matrix, see also length
sqrt	Square root
ss	Create state-space models or convert LTI model to state space, see also tf

ssdata	Access to state-space data. See also tfdata
stairs	Stairstep plot for discrete response
step	Plot the step response, see also impulse, lsim
subplot	Divide the plot window up into pieces, see also plot, figure
text	Add a piece of text to the current plot, see also title, xlabel, ylabel, gtext
tf	Creation of transfer functions or conversion to transfer function, see also ss
tfdata	Access to transfer function data, see also ssdata
title	Add a title to the current plot
wbw	Returns the bandwidth frequency given the damping ratio and the rise or settling time.
xlabel/ylabel	Add a label to the horizontal/vertical axis of the current plot, see also title, text, gtext
zeros	Returns a vector or matrix of zeros
zgrid	Generates grid lines of constant damping ratio (zeta) and natural frequency (Wn), see also sgrid, sigrid

How to use these commands?

For each command go to MATLAB command prompt and write **help (corresponding command).** After that, follow the given information and try to make yourself handy with that command using that example.

Let's have an example-

i. Suppose, we want to know about the command **rloucs.** So write **help rlocus** and press enter.
ii. After the command a description will be given like below:

```
>> help rlocus
 RLOCUS  Evans root locus.

    RLOCUS(SYS) computes and plots the root locus of the single-input,
    single-output LTI model SYS.  The root locus plot is used to
    analyze the negative feedback loop

                      +-----+
          ---->O----->| SYS |----+---->
              -|      +-----+    |
               |                 |
               |       +---+     |
               +-------| K |<----+
                       +---+

    and shows the trajectories of the closed-loop poles when the feedback
    gain K varies from 0 to Inf.  RLOCUS automatically generates a set of
    positive gain values that produce a smooth plot.

    RLOCUS(SYS,K) uses a user-specified vector K of gain values.

    RLOCUS(SYS1,SYS2,...) draws the root loci of multiple LTI models
    SYS1, SYS2,... on a single plot.  You can specify a color, line style,
    and marker for each model, as in
       rlocus(sys1,'r',sys2,'y:',sys3,'gx').

    [R,K] = RLOCUS(SYS) or R = RLOCUS(SYS,K) returns the matrix R of
    complex root locations for the gains K.  R has LENGTH(K) columns
    and its j-th column lists the closed-loop roots for the gain K(j).

    See also sisotool, POLE, ISSISO, ltimodels.
```

iii. Using this information practice that example and try other commands.

Lab Task-1

Prepare a sheet where at least 20 MATLAB command you have practiced in the way like-

i. Write the syntax.
ii. Example how to use that command
iii. Executed command output/waveshape.

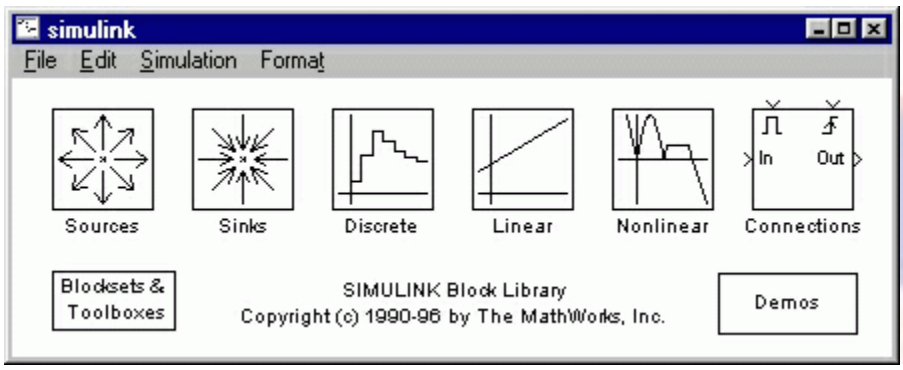
Objective-2: **Getting used to with Simulink blocks for system development.**

Simulink Basics Tutorial - Block Libraries

List of blocks to describe:

1. Sources
2. Sinks
3. Discrete
4. Linear
5. Nonlinear
6. Connections

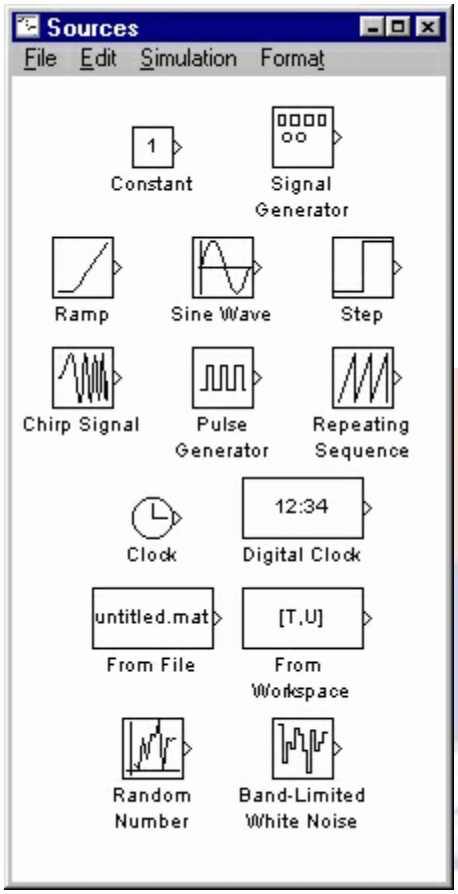
Simulink contains a large number of blocks from which models can be build. These blocks are arranged in **Block Libraries** which are accessed in the main Simulink window shown below



Each icon in the main Simulink window can be double clicked to bring up the corresponding block library. Blocks in each library can then be dragged into a model window to build a model.

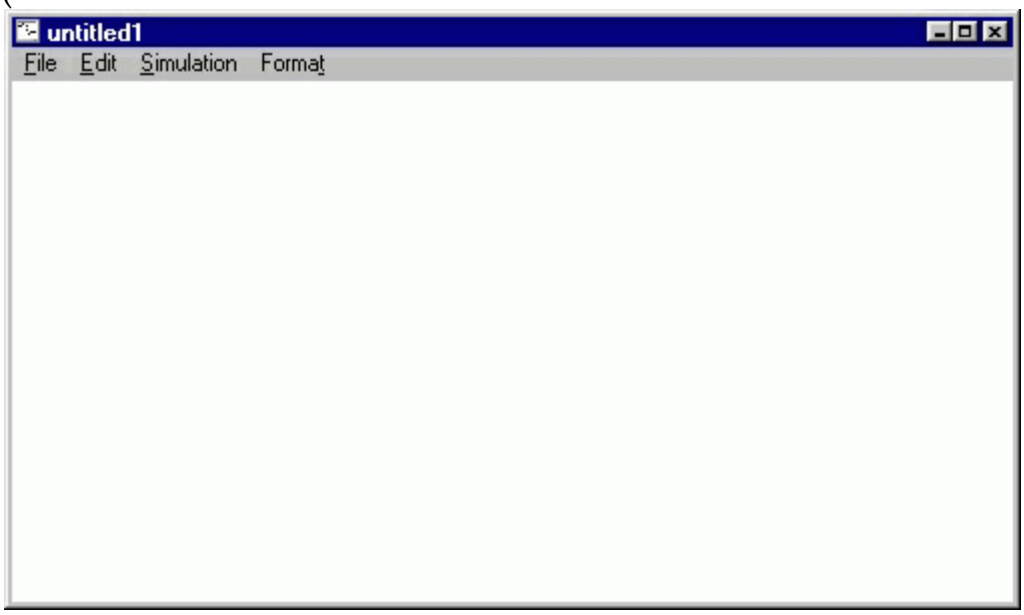
1. Sources

Source Blocks are used to generate signals. Double-click on the **Sources** icon in the main Simulink window to bring up the Sources window.



Notice that all of the source blocks have a single output and no inputs. While parameters in each of these blocks in the library can be modified by double-clicking the block, it is best to not modify the blocks until they have been copied into a model window.

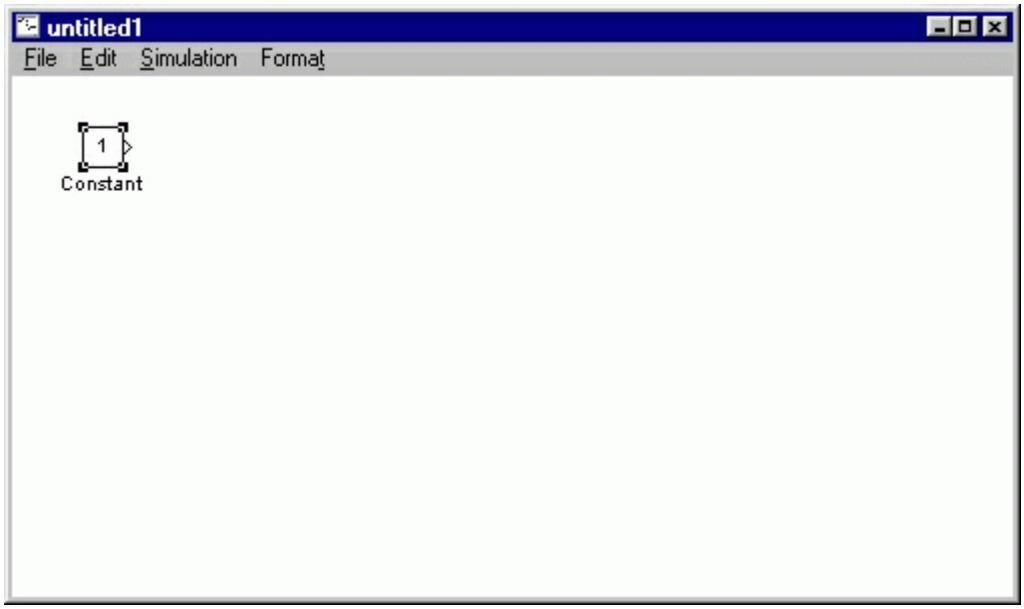
In order to examine these blocks, create a new model window (select **New** from the **File** menu in the Simulink window or hit Ctrl+N).



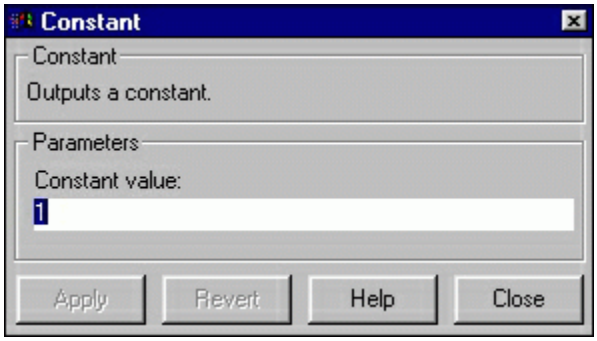
Constant

The Constant Source Block simply generates a constant signal. The constant output value is displayed in the middle of the block, with a default value of 1.

To use this block, drag it from the Sources window into your new model window.



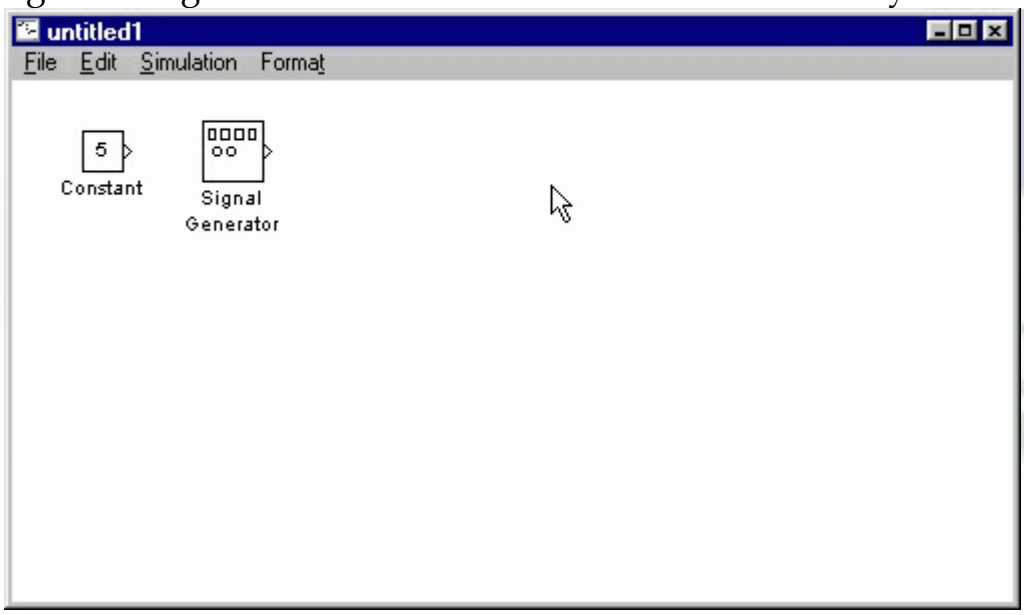
To change the constant output value, double-click on the block in your model window to bring up the following dialog box.



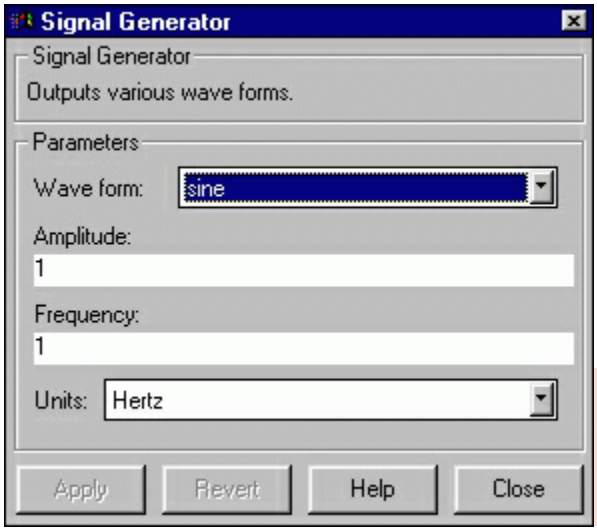
Change the constant value field from 1 to some other value, say, 5, and close the dialog box. Your model window will reflect the update by displaying a 5 in the middle of the constant block.

Signal Generator

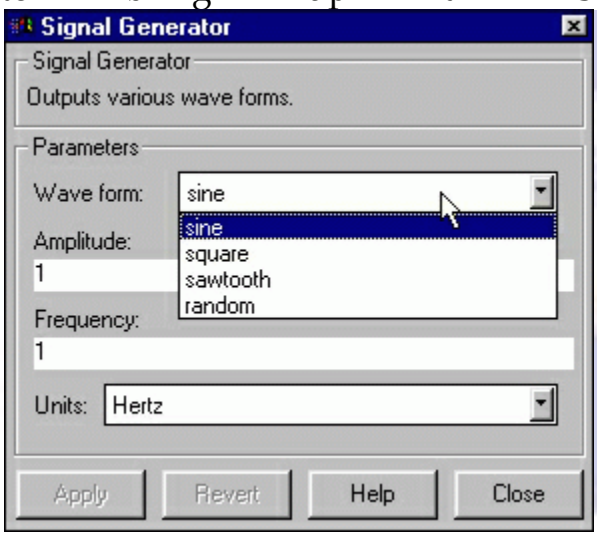
The Signal Generator Source Block is a general-purpose source which encompasses some of the other blocks' functions. It generates periodic waveforms such as sine, square, and sawtooth waves as well as a random signal. Drag this block from the Sources window to your model window.



By default, the Signal Generator generates a sine wave with an amplitude of 1 and a frequency of 1 Hz. To change this, double-click the Signal Generator in your model window to bring up the following dialog box.



The Amplitude and Frequency can be changed in this dialog box, as well as the type of waveform. To change the waveform, click on the Waveform field to bring up a list of possible waveforms.



The desired waveform can be selected from this list.

Ramp

The Ramp Source Block generates a signal which is initially constant and begins to increase (or decrease) at a constant rate at a specified time. The slope, start time, and initial output can be specified.

Sine Wave

The Sine Wave Source Block generates a sinusoidal signal. The Amplitude and Frequency can be specified, as well as the Phase (unlike the Signal Generator). There is a fourth parameter, the Sample Time, which can be used

to force the Sine Wave Source to operate in discrete-time mode (more about Discrete Time systems in Simulink later.)

Step

As described earlier, the Step Source Block generates a step function. The initial and final values can be specified, as well as the step time.

Chirp Signal

The Chirp Signal Source Block generates a sinusoidal signal which scans over a range of frequencies. The initial and final frequencies as well as the scan time can be specified. The amplitude is always 1, and the chirp signal repeats itself after each frequency scan.

Pulse Generator

The Pulse Generator Source Block generates a pulse train of varying duty cycle. The signal switches between 0 and the specified value starting at a particular time. The Period, Duty Cycle, Amplitude, and Start Time can be specified.

Repeating Sequence

An arbitrary set of points (t,y) can be specified. These points are entered as a vector specifying the time values, and a vector specifying the corresponding output values at those times. The output is linearly interpolated between the specified time values. At the last time value, the output immediately starts over, possibly with a discontinuous transition.

Clock

The Clock Source Block generates a signal equal to the current time in the simulation. This is useful when the output of a simulation is exported to MATLAB but occurs at uneven time steps. The clock's output reflects the times at which the other signals outputs occur.

Digital Clock

The Digital Clock Source Block generates a strictly periodic time signal at a specified sampling interval.

From File

The From File Source Block outputs a signal taken from a specified .mat file. A matrix saved in MATLAB as a .mat file will become a signal where the first row of the matrix specifies the time values. This is similar to the Repeating Sequence Source Block.

From Workspace

The From Workspace Source Block is identical to the From File Source Block except the values are taken from a variable (or expression) in the MATLAB Workspace.

Random Number

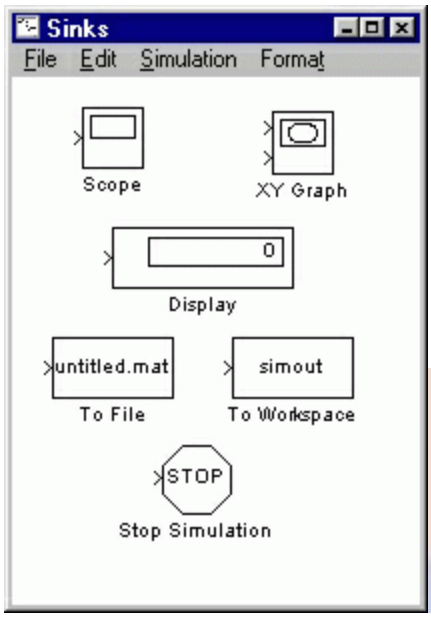
The Random Number Source Block generates a sequence of random numbers generated with the specified random number seed. Because of the seed, the same sequence can be applied to more than one simulation.

Band-Limited White Noise

The Band-Limited White Noise Source Block generates a random signal which changes at a specified sample period. The strength of the signal and a random number seed can also be specified.

2. Sinks

Sink Blocks are used to display or output signals. Double-click on the **Sinks** icon in the main Simulink window to bring up the Sinks window.



Notice that all of the sink blocks have inputs and no outputs. Most have a single input.

Scope

The Scope Sink Block was described earlier. It is used to display a signal as a function of time.

XY Graph

The XY Graph Sink Block plots one signal against another. It is useful for phase-plane plots, etc.

Display

The Display Sink Block is a digital readout of a signal at the current simulation time.

To File

The To File Sink Block saves a signal to a .mat file in the same way that the From File Source Block reads from a file. The sampling time can be specified, but is not necessary.

To Workspace

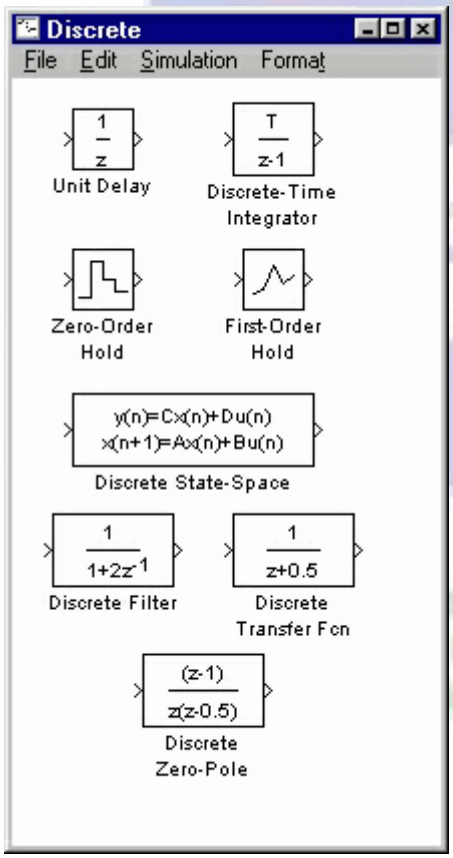
The To Workspace Sink Block stores a signal in a specified workspace variable. Unlike the To File Sink Block, the time is not saved in the variable, and must be stored separately.

Stop Simulation

This is a special control block which is triggered to stop the current simulation when its input is non-zero.

Discrete

Discrete Blocks are elements of discrete time dynamic systems. Double-click on the **Discrete** icon in the main Simulink window to bring up the Discrete window.



In order for these block to interact with continuous time blocks (sources and sinks, for example) the sample time can be specified in all of the Discrete Blocks.

Unit Delay

The Unit Delay's output is equal to the input delayed by one sample time.

Discrete-Time Integrator

This is the discrete time approximation of a continuous-time integrator. The approximation method can be specified as well as the initial condition and saturation limits.

Zero-Order Hold

Outputs a stepwise-constant version of the input signal with a specified sampling period.

First-Order Hold

Outputs a piecewise-linear version of the input signal with a specified sampling period.

Discrete State-Space

This is a discrete-time dynamic system in state-space form. A, B, C, and D matrices can be specified, as well as initial conditions.

Discrete Filter

This is a discrete-time filter in rational function form. Vectors containing coefficients of the polynomials in z^-1 are specified.

Discrete Transfer Fcn

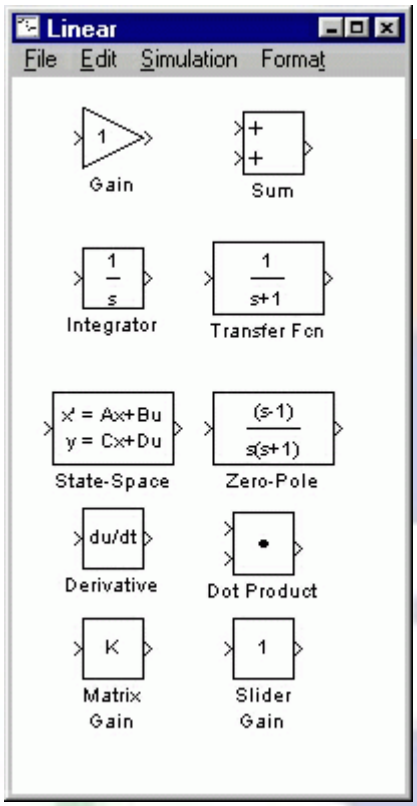
This is the standard form of a SISO LTI discrete time system. The transfer function polynomials are represented as coefficient vectors in terms of z.

Discrete Zero-Pole

A discrete-time transfer function can be represented as list of z-plane poles and zeros. The gain can also be specified.

3. Linear

Linear Blocks are elements of linear continuous-time dynamic systems. Double-click on the **Linear** icon in the main Simulink window to bring up the Linear window.



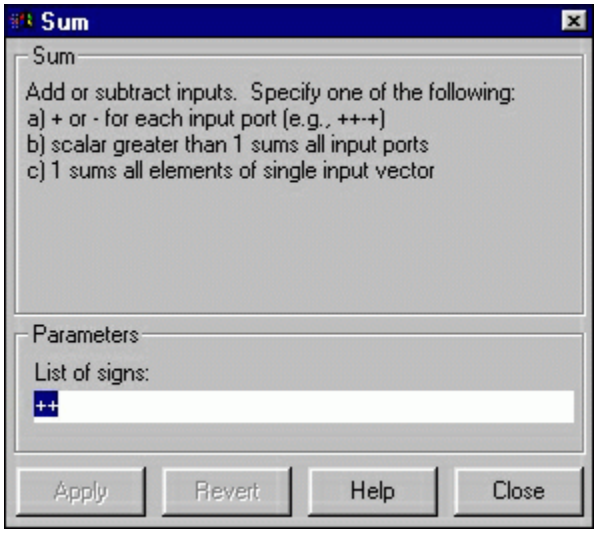
Gain

This is a scalar or vector gain. The specified gain multiplies the input. The output is either a scalar or vector signal following normal vector-scalar multiplication rules.

Sum

The Sum Block adds (or subtracts) two (or more) signals and outputs their sum (or difference). The two inputs must either all be scalars, or all be vectors of the same dimension. The output is the same dimension as the inputs.

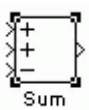
Drag this block into your model window and double click on it, bringing up the following dialog box.



By default, the Sum block adds two signals. The list of signs field represents both the number of inputs and whether to add or subtract them. To make the Sum block add two signals and subtract a third, change the list of signs to the following:

++-

Each element in the list corresponds to one of the signals. Close the dialog box. The Sum block in your model window should now have three inputs, one of which has a minus sign as shown below.



Integrator

The output of the Integrator is the integral of the input. An initial condition can be specified, as well as saturation limits. This block is very useful for modeling systems.

Transfer Function

Numerator and denominator polynomials can be specified to create a standard SISO LTI system transfer function.

State Space

A, B, C, and D matrices can be specified to create a LTI state space system. Inputs and outputs may be vector signals depending on the sizes of the matrices.

Zero-Pole

Vector lists of zeros and poles can be specified to create a transfer function. DC gain is also specified.

Derivative

The output is equal to the derivative of the input.

Dot Product

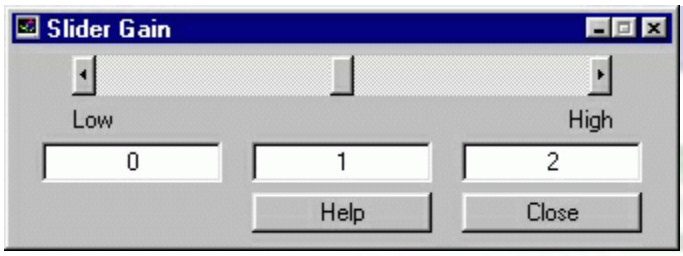
The output is equal to the dot product of two vector signals.

Matrix Gain

The output is equal to the input times a specified constant matrix. The size of the input and output vector signals must match the size of the matrix.

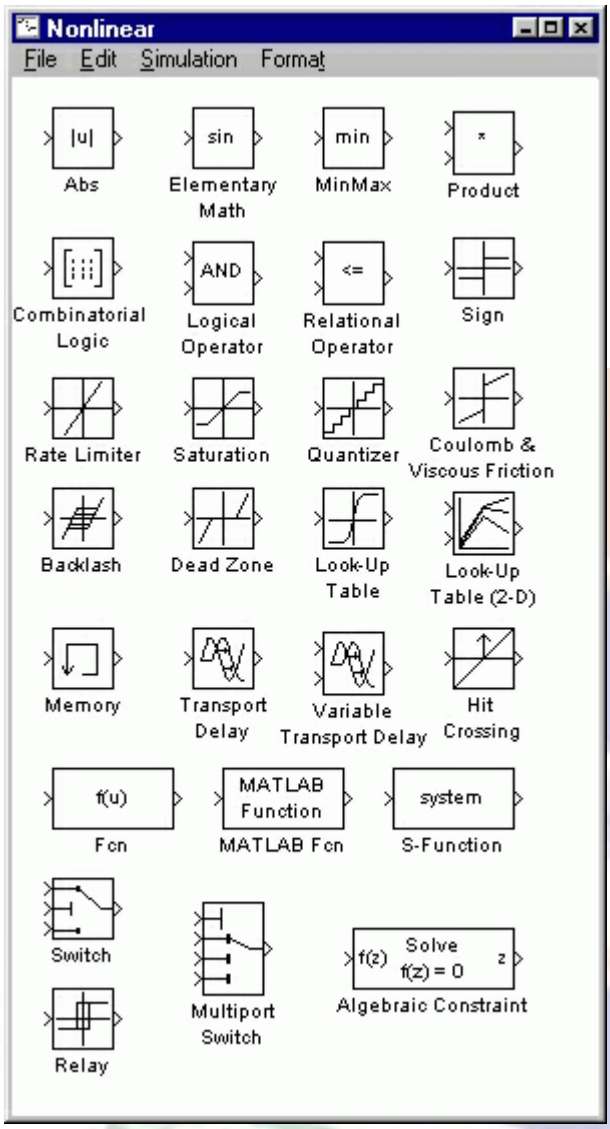
Slider Gain

This multiplies the input by a scalar constant which is specified by moving a slider on the screen as shown below. The limits of the slider can be specified.



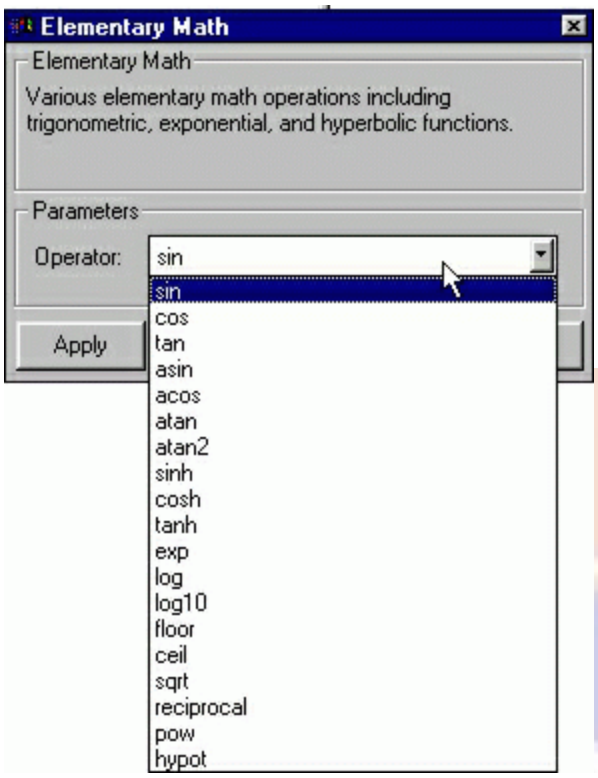
4. Nonlinear

Nonlinear Blocks are elements of nonlinear continuous-time dynamic systems. Most of these have special-purpose applications and will not be used in the tutorials. Only the most relevant Nonlinear blocks will be discussed here. Double-click on the **Nonlinear** icon in the main Simulink window to bring up the Nonlinear window.



Elementary Math

The output of this block is a basic mathematical function applied to the input. Copy this block into your model window and double click on it. The resulting dialog box has one field, which when clicked on, give the following menu of functions.



The desired function can be selected from the list, and the choice will be displayed in the block in your model window.

Product

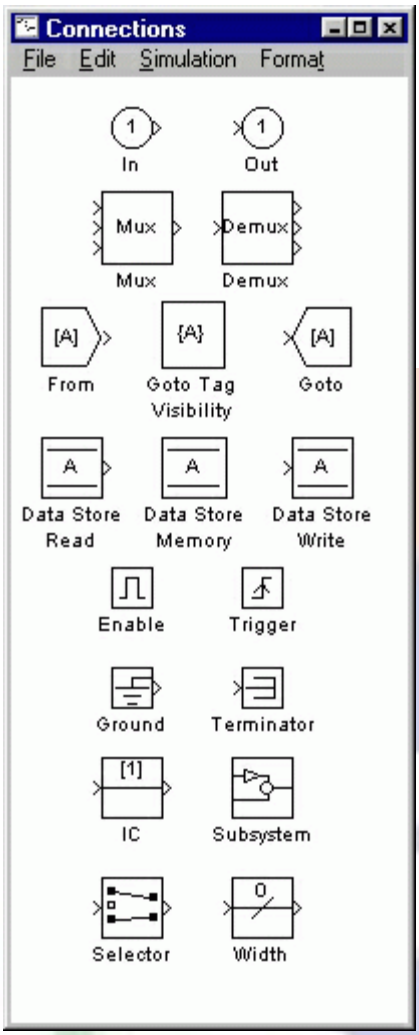
The output is equal to the product of the inputs. The number of inputs can be specified.

Fcn

Arbitrary MATLAB expressions can be represented by this block.

5. Connections

Connection Blocks are used to organize and combine signals and systems. These also have special purpose applications, and only some of these will be described here. Double-click on the **Connections** icon in the main Simulink window to bring up the Connections window.



Subsystem, In, Out

The Subsystem block is used as a system "macro", where one Subsystem block can be used to represent an entire set of blocks. When double-clicked, the subsystem block brings up a blank model window. Note that the Subsystem block has no inputs or outputs. For each input a Subsystem has, an In block is used. For each output a Subsystem has, an Out block is used. For more information on Subsystems and In and Out blocks, see the Interaction with MATLAB tutorial page. For an example of the creation of a subsystem, see the DC Motor Speed Modeling in Simulink example (as well as other examples.)

Mux, Demux

The Mux (Multiplexer) block is used to combine two or more scalar signals into a single vector signal. Similarly, a Demux (Demultiplexer) block breaks a vector signal into scalar signal components. The number of vector components must be specified in each case. For an example of the use of a Mux block see the Bus Suspension Modeling in Simulink example.

Report preparation

Please prepare your report with yourself & don't try to copy. Report should cover the following Topics:

1. Objective
2. Apparatus/Utility software:
3. Description of Task:
4. MATLAB code & Result:
5. Analysis of Each output:
6. Discussion: